

QEM-6: Mirror-Flip Quantum Unit as the Minimal Structural Cell of Entanglement

Author: Youfu Wang (王佑福)

ORCID: 0009-0005-5998-6024

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Abstract (EN):

We introduce the Mirror-Flip Unit (MFU) as the minimal structural cell that realizes and stabilizes multi-body entanglement in the QEM framework. The MFU is a dual-layer mirror lattice supporting Σ/Δ collective modes with a controllable Δ -domain-wall profile that confines coherence. We derive governing equations near stationary textures, obtain the scaling $\xi \sim |\theta - \theta_c|^{-1/2}$, and predict falsifiable signatures: π -echo, sech^2 standing envelopes, and a causal front bounded by $v_{\text{max}} \approx (J \xi)/\hbar$. Protocols are provided for wall-fit (ξ), front-tracking (v_{max}), and stability-window mapping.

摘要 (ZH):

本文在 QEM 框架下引入“镜像翻转单元 (MFU)”作为实现并稳定多体纠缠的最小结构单元。MFU 由支持 Σ/Δ 集体模式的双层镜像晶格构成，其可通过可控的 Δ -畴壁轮廓限制相干性。我们推导出近静态纹理下的 governing equations，得到标度律 $\xi \sim |\theta - \theta_c|^{-1/2}$ ，并预测可证伪的特征： π -echo、 sech^2 驻波包络以及由 $v_{\text{max}} \approx (J \xi)/\hbar$ 界定的因果前沿。本文提供了壁拟合 (ξ)、前沿追踪 (v_{max}) 及稳定性窗口映射的协议。

Series note: This work is part of the Quantum Entanglement Mechanics (QEM) series by Youfu Wang (2025).

1. Introduction

QEM-1-5 established layered mirror lattices, domain-wall stabilized textures (QEM-4), and a causal front limiting flip-wave propagation (QEM-5). QEM-6 identifies the Mirror-Flip Unit (MFU) as a portable device-level cell that encapsulates these effects for scaling to arrays and informs chip-oriented timing and fan-out constraints.

2. MFU Definition and Geometry (Σ/Δ modes)

MFU Geometry: A four-node dual-layer loop A-B-A'-B' with mirror exchange κ and intra-layer coupling J. Σ/Δ collective modes govern symmetric/antisymmetric phase dynamics. A stationary Δ -domain wall threads the cell and confines coherence.

Parameters: lattice spacing a , mirror spacing d , operating bias θ (relative to θ_c), dissipation γ .

3. Governing Equations and Stationary Texture

Coarse-grained phase $\phi(x,t)$ and coherence amplitude ψ obey a coupled-mode description. Linearization around a stationary wall yields a ϕ^4 -type Δ -equation with kink (tanh) solution; Σ provides restoring stiffness and sets effective sound speed c_{eff} . Near the resonant operation, standing textures exhibit $\sim \text{sech}^2$ envelopes.

Schematic: $\partial_t^2 \phi - c_{\text{eff}}^2 \partial_x^2 \phi + \alpha \phi - \beta \phi^3 = S_{\Delta}(x,t)$, with $\alpha, \beta = \alpha(J, \kappa, \theta), \beta(J, \kappa, \theta)$.

4. Stability Window and Scaling

Scaling: $\xi \sim |\theta - \theta_c|^{-1/2}$. The causal front bound from QEM-5 reads $v_{\max} \approx (J \xi)/\hbar$. Tuning ξ and J sets the MFU's timing budget and safe fan-out. Predicted signatures: (i) π -echo; (ii) sech^2 envelope; (iii) bounded front velocity.

5. Experimental Protocols

- (1) Wall-fit ξ : prepare a Δ -kink; fit tanh profile to extract ξ .
- (2) Echo π -flip: mirror-symmetric drive; detect echo-phase inversion $\approx \pi$.
- (3) Front-tracking v_{max} : launch a local perturbation; track arrival; linear fit gives v_{max} .
- (4) Noise sweep: vary θ, γ to map texture survivability; delineate stable window.

6. Discussion

MFU operationalizes entanglement mechanics into a minimal device cell. It is the turning layer from abstract dynamics to hardware mapping. By separating universal structure (public) from engineering parameters (internal), QEM-6 enables reproducible verification and chip-level synthesis.

7. Related Work

Correlation spreading in non-relativistic many-body systems is constrained by Lieb–Robinson-type bounds establishing finite light-cone-like velocities. Experiments in cold-atom optical lattices resolved ballistic correlation fronts after a quench. The Calabrese–Cardy framework provides a quasiparticle picture for post-quench entanglement growth. On the structural side, ϕ^4 kink (tanh) and Pöschl–Teller (sech^2) profiles supply canonical texture shapes. QEM-6 recasts these ingredients into a hierarchical mirror-lattice architecture and identifies the flip-wave as the minimal operational unit with device-oriented protocols.

8. Conclusion

QEM-6 establishes the Mirror-Flip Unit as the minimal structural cell of entanglement. The ξ -scaling and the causal front bound connect structure, stability, and dynamics in a single unit, paving a practical path toward QEM-7 energy-field modeling and QEM-8 unification.

References

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